# (U) ModelNameGoesHere
# Test Description Document




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(U) FOR TITLE PAGE
(U) TIMELINE: MOST ITEMS: Ready by SRR, finalized by PDR. REVISION NUMBERS & DATES: Finalized at FDR.
(U) "Prepared By" and "Classified By" must be a company name, not an individual's name
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(U) "Distribution/Dissemination Controls" line is only needed if the document contains CUI material

(U) FOR REVISION TABLE:
(U) TIMELINE: Each milestone submission needs its own row, in 0.# format. Final delivery is 1.0. Dates are YYYYMMDD.

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| Revision | Date | Author | Description of Changes |
|---|---|---|---|
| 1.0 | 201609 | NGIC (Everett) | Initial Draft |
| 2.0 | 202304 | NGIC (Everett) | Revised for new Projects |
| 2.1 | 20241014 | BAH | Revised for new projects and formatting |
| 2.2 | 20241229 | BAH | Revised based on feedback |
| 2.3 | 20250109 | BAH | Revised based on feedback |
| 3.0 | 20250217 | BAH | Baselined to 3.0 release for document templates |
| 3.1 | 20250729 | BAH | Draft of 4.0 templates |
| 4.0 | 20251114 | BAH | Baselined to 4.0 release for document templates |

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(U) HIGH LEVEL GUIDANCE FOR THIS DOCUMENT:
(U) Please maintain consistency across the document within a test family and across the document as much as possible. Maintain consistent labels, naming conventions, and output displays to include axis, scales, etc. This allows the reader ease in comparing data from one plot to another.
(U) This does require diligence and extra effort but must be emphasized.
(U) Example: If OJX-1 is the label in the laydown then that is the name that should be used through out. If the Overarching Dynamic Electronic-Warfare System Standard Architecture (ODESSA) asset number or tag is "6" in the provided scenario file there should be a note in the text to say that.
(U) If this type of information can be standardized for the entire set of tests then it is recommended. Include in a high level table if it makes sense.
(U) NOTE: It is recommended that naming that is used considers something that can be used both unclassified and classified or done in a way that is easy to update in the secure space.

# 1 (U) Introduction

(U) This document summarizes the test plan for the ModelNameGoesHere Threat Modeling and Analysis Program (TMAP) model developed by Booz Allen Hamilton (BAH) in the [architecture name here] (ex: ODESSA) for National Ground Intelligence Center (NGIC). The purpose of this Test Description Document (TDD) is to define a strategy for verification of the model implementation. Verification is the process of determining the model and associated data accurately represent the conceptual description and specification as described in the model requirements. Validation is the determination that the model is an accurate representation of the real world from the perspective of the intended use and will be performed by NGIC.

(U) The ModelNameGoesHere TDD documents the testing suite for the ModelNameGoesHere *type* model developed by BAH in the ODESSA framework in support of the NGIC.

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# 2 (U) Requirements Verification Traceability Matrix

(U) TIMELINE: Test plan should be near final draft (all tests planned out with behavior tested at a minimum) for PDR for review by NGIC and finalized by CDR.
(U) The test plan consists of the test framework and the linking of test cases to requirements. Most requirements require testing and thus have an associated KPIs; some requirements can be confirmed by sbservation.
(U) The table below should include all of the requirements and corresponding tests which will demonstrate that the requirement is met. The test coverage column should include "ST1" for system family test one. If a unit test covers that, then use "unit test" in that column with a unit test identifier. If unit tests are used to verify a requirement, then a section should be included in the TDD that describes it/them.
(U) Test results/status will be updated for Test Readiness Review (TRR) and should include the following options as the text:
(U) Pass – The test case ran successfully as written
(U) Pass with C – Although a particular test step failed, the test case will be marked as pass with caveat because the deficiencies do not substantially impair the required performance. NGIC Technical Point of Contact (TPOC) should be notified immediately if this is the expect result to gain concurrence.
(U) Fail – Unable to complete the test or a requirement could not be verified
(U) Pending – Step failed but needs analysis, after which, disposition is changed as identified above. This is only an option while in development for incremental updates at milestones. By Final Design Review (FDR), its should be one of the above options.

Table 1: (U) Requirements Verification Traceability Matrix

| UNCLASSIFIED | | | |
|---|---|---|---|
| Requirement | Description | Test Coverage | Test Result / Status |
| 3: General System Model | | | |
| 3.1 | Description | ST1 | Not Tested |
| 4: Antenna Parameters | | | |
| 4.1 | Description | ST1 | Not Tested |
| 5: RF Performance Parameters - Frequency, Range, Power | | | |
| 5.1 | Description | ST1 | Not Tested |
| 6: Timing and Scheduler | | | |
| 6.1 | Description | ST1 | Not Tested |
| 7: Modes and Logic | | | |
| 7.1 | Description | ST1 | Not Tested |
| 8: Sensors and Weapon Accuracy | | | |
| 8.1 | N/A | N/A | N/A |
| 9: Environment and Geometric Nuances | | | |
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Table 1: (U) Requirements Verification Traceability Matrix (Continued)

| UNCLASSIFIED | | | |
|---|---|---|---|
| Requirement | Description | Test Coverage | Test Result / Status |
| 9.1 | Description | ST1 | Not Tested |
| 10: Target Classes and Target Quantity | | | |
| 10.1 | Description | ST1 | Not Tested |
| 11: Digital Signal Processing and Waveforms | | | |
| 11.1 | Description | ST1 | Not Tested |
| 12: Electromagnetic Attack (EA) Capability | | | |
| 12.1 | Description | UT1 | Passed |
| 13: Electromagnetic Protection (EP) Capability | | | |
| 13.1 | Description | ST1 | Not Tested |
| 14: Tactics, Techniques, and Procedures | | | |
| 14.1 | Description | ST1 | Not Tested |
| 15: Propagation | | | |
| 15.1 | Description | ST1 | Not Tested |
| 16: Model Outputs | | | |
| 16.1 | Description | ST1 | Not Tested |
| | | | UNCLASSIFIED |

# 3 (U) Test Structure

## 3.1 (U) Tests Description

(U) TIMELINE: Should be started by PDR and finalized at CDR.
(U) This section should generally not be modified. However, if you have a good reason to deviate from this standard, your model may modify this format to better suite your testing needs with approval from, at a minimum; the model lead, the NGIC M&S TPOC, and the analyst for the system.

(U) The tests for the ModelNameGoesHere model are split into a number of test families which represent multiple geometries or over-arching test conditions, and these test families contain test categories that demonstrate multiple requirements. Miscellaneous test categories that include tests that don't fulfill any requirements but show useful results, are also included. Each family is split into a series of tests, which will fulfill all of the requirements for the corresponding requirement set.

(U) System tests assume that all unit tests pass for each component used in the system test.

(U) Fully completed tests will be marked with a ✓, and their results will be explored in more detail below.

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Each Test Case describes the Key Performance Indicator (KPI)s that test assesses. A table of all KPIs and the associated equations appears in the User and Analyst Manual (UAM) as Table #.

### 3.2 (U) Test Numbering Scheme

(U) TIMELINE: Should be started by PDR and finalized at CDR.
(U) Briefly state the schema used to number and delineate test cases.

(U) System tests will be named according to a set of major Test Families which will be described below using the designation "TF1" to mean "Test Family 1". Some requirements will take more than a single test to meet, and some tests can be designed to satisfy more than one requirement, so a 1-1 tests-requirements match-up will not exist. Please refer to the requirements traceability matrix in section 2 for more details on tests to requirements mapping for model requirements.

## 4 (U) Model Testing

### 4.1 (U) Scopes of Test Cases

(U) **Table 2** summarizes each test case's purpose and lists which components affect the outcome of the test. **Table 2** shows how this table would appear in OD5's TDD.

(U) TIMELINE: Should be started by PDR and finalized at CDR.
(U) Each test case has its own row. Briefly state the purpose of the test case. List which components participate in the test.
Example: (U) suppose "Test 3" ensures that the tracking logic calculations are correct. Test 3 may use runtime position information from the model's receiver, or it may use canned data imported from a file. In the first case, the receiver would be listed as a dependent component because it has to work correctly for Test 3 to pass. The receiver is not a dependent component in the second case because Test 3 would pass even if the receiver is broken. One purpose of this column is to show whether Test 3 needs to be rerun if the receiver changes somehow later, after Test 3 was run the first time.

Table 2: (U) System Testing Prerequisites and Scope

| UNCLASSIFIED | | |
|---|---|---|
| Test Case | Summary | Dependent Components |
| 1.1 | Verifies frequency, PRIs, and PWs of transmitted SPDWs | Radar Transmitter |
| 1.2 | Verifies that antenna patterns are correctly applied to received SPDWs | Antenna |
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Table 2: (U) System Testing Prerequisites and Scope (Continued)

| UNCLASSIFIED | | |
|---|---|---|
| Test Case | Summary | Dependent Components |
| 2.1 | Verifies coverage of each sector scan mode | ScanPattern |
| 3.1 | Verifies each radar mode returns the correct information | RadarMode |
| 3.2 | Verifies that tracking logic returns the correct information | TrackManager KalmanFilter |
| 4.1 | Verifies that tracks are processed into correct POO and POI calculations | KalmanFilter TrackManager |
| 5.1 | Verifies that the model correctly handles atmospheric loss and terrain data | Environment (ext) SpdwManager CFAR |
| 6.1 | Verifies that targets are correctly classified as ballistic or non-ballistic | TrackManager KalmanFilter |
| 6.2 | Verifies that up to five simultaneous targets are processed correctly | TrackManager KalmanFilter |
| 7.1 | Verifies the waveform characteristics of every mode | SpdwManager |
| 9.1 | Verifies uplink and downlink with an external C2 | KalmanFilter |
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## 4.2 (U) Test Family X: Descriptive Name

(U) TIMELINE: Should be started by PDR and finalized at CDR.
(U) REQUIREMENTS
(U) **Hierarchy** Test Families are comprised of Test Cases. EXAMPLE: Test Family X consists of Test Cases X.1, X.2, and X.2a.
(U) **Representative, not Exhaustive** The test family needs to be thoroughly tested in the code, but the TDD need not have a separate Test Case for every coded test.
(U) This overarching section should be a brief description of the purpose and function of the test family. Defer discussion of shared characteristics and default inputs to the subsections below.
(U) EXAMPLE: Scenario X.2b also belongs in this Test Family, but is not significantly different from Scenario X.2a, and so is only briefly included in further detail in a similar test case.

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### 4.2.1 (U) Test Cases' Shared Characteristics

(U) TIMELINE: Should be started by PDR and finalized at CDR.
(U) REQUIREMENT **Shared Characteristics** Each Test Family is defined by some shared similarity. For instance, all tests in this family attempt to jam a search radar, or all test in this family share the same geometry.
(U) EXAMPLE: These tests all disable the TER, and focus on TAR functionality.
(U) EXAMPLE: These tests all focus on varying jamming modes, and keep the rest of the model parameters relatively static.
(U) EXAMPLE: These tests all use a geometry of interest provided by NGIC.

### 4.2.2 (U) Test Family Default Inputs

REQUIREMENT
(U) TIMELINE: Should be started by CDR and finalized at CDR.
(U) **Parameters** Each Family has in its introductory section a table of parameters that define its base parameters, which are used by its Cases. This table lists default values. Each Case has a similar table listing that Case's unique values.
(U) **Scope** This table lists all parameters that *both* differ from the Table of Default Parameters in the UAM *and* are the same for every test case in this family.
(U) **Content** List that parameter's default value for Cases in this Family. Each individual Case will have its own table of values unique to that test.

(U) **Table 3** shows those parameter settings which differ from the defaults in Table # of the ModelNameGoesHere Users and Analyst's Manual and are also identical for all tests in this test family. Each Test Case has a table of parameters for which that test case uses a unique value.

Table 3: (U) Sample Test Family: Family-Wide Parameter Settings

| **UNCLASSIFIED** | | |
|---|---|---|
| Input Name | Description | Value(s) |
| Simulation Start Date | Start Date for the Simulation | 07/10/2024 |
| | | **UNCLASSIFIED** |

### 4.2.3 (U) Test Case X1: Descriptive Name

(U) TIMELINE: Should be finalized by Critical Design Review (CDR).
(U) REQUIRED: Brief prose description of the test and laydown figure. ENCOURAGED: explanatory figures and/or tables.

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106 #### 4.2.3.1 (U) Behavior Tested

(U) TIMELINE: Should be finalized by CDR.
(U) REQUIRED: Prose description of the behavior tested. Identify the items in the requirements traceability matrix that are assessed by this test. ENCOURAGED: explanatory figures and/or tables.
(U) Basic example: Test Case 1 examines the functionality of the Early Warning Receiver (EWR), without a jammer present. The test centers around antenna parameters such as directionality, receiver frequency, and mainbeam power. The requirements associated with Test Case 1 are listed in table below.
(U) Additional example: Test Case 1 examines the model's ability to apply appropriate FOV limits to jamming commands and beam steering; as well as interpolating the gain according to changing geometry.
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108 (U) Behavior Tested goes here.

(U) TIMELINE: Should be finalized by CDR.
(U) REQUIRED: There must be a figure showing the basic geometry setup of the test case. If the test family has a shared geometry, then this figure may be moved to the test family section. The figure below demonstrates a way of plotting the geometry. Make sure to include labels on both axis as well as legends. Make sure it's clear which direction the platform is moving.
(U) In addition to the figure, please include a written statement describing the geometry of the test case.
(U) Example: For each of the test cases in Test Family 1, the OJX-1 is located [LLA] (as denoted by red dot in figure 1) and is facing [insert cardinal direction and/or az/el]. It is set to the test specific parameters in table 5 below. The Straight and Level Platform (SLP) starts the scenario located at [LLA] approximately [y-distance]km [cardinal direction and az/el] from OJX-1. The SLP is traveling [cardinal direction and az] at Z- m/s which is a [direct approach, crossing route, etc.] compared to the stationary OJX-1..
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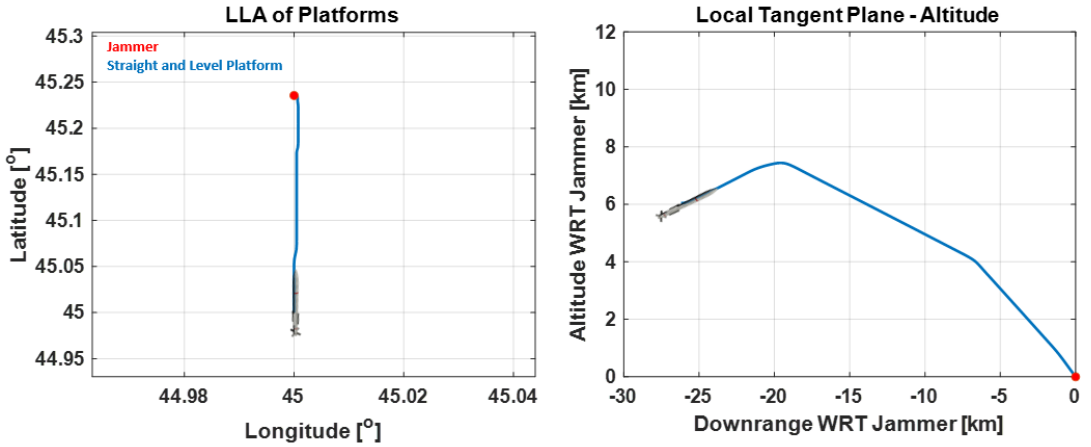



**Figure 1: (U) Test Case 1 Geometry**

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Table 4: (U) Requirements Traceability Matrix

| UNCLASSIFIED | | | |
|---|---|---|---|
| Requirement | Description | Test Coverage | Test Result / Status |
| 3: General System Model | | | |
| 3.1 | Description | 1.1 | Not Tested |
| | | | UNCLASSIFIED |

#### 4.2.3.2 (U) Prerequisites

(U) TIMELINE: Should be finalized by CDR.
(U) REQUIREMENT Identify what classes, subsystems, etc. that must be completed and tested before this test can be concluded. If this is a system test or simulation environment test, identify what other models or federates are required. One of the purposes of this section is to improve traceability so that if a "prerequisite" class or subsystem changes we can easily identify which downstream tests need to be re-run or re-evaluated; or if this test is anomalous the evaluator can more easily identify the upstream tests to inspect. Another purpose of this section is to understand scheduling considerations for development. Another purpose of this section is to facilitate addition of stop conditions or warnings in test automation or review. For example, if a "prerequisite" test fails or its results are out of bounds, then downstream tests should be stopped or flagged because we know the downstream tests will provide erroneous results. To this end, the table below has Primary, Secondary, and Unused component sections for each part of the test case, as well as the overall test case. Changes to Primary and Secondary components are guaranteed and likely, respectively, to at least require re-inspection of the test case. Unused components are components that are either intentionally disabled or designed to not function during this test case, and would not require re-inspection of the test case if they were changed. Any component that does not fall into one of these categories is generally assumed to be proven functional through unit tests and other test cases, and likely to not affect this test case.
(U) Basic example: Test Case 3 requires that all of the radar modes are functioning properly, as these components are the primary focus of this test case. This test case also requires that the radar processor is functionally processing signals and moving between modes where applicable.

(U) **Table 5** shows which components are assessed by this Test Case. Primary Components relate directly to the behavior tested and expected results. Secondary Components are active during the test. Unused Components are those turned off or simulated with truth values or lookup tables. Any changes made to Primary or Secondary Components or to lookup tables used by them will require that this test be rerun.

Table 5: (U) Test Case Prerequisites

| UNCLASSIFIED | | | | |
|---|---|---|---|---|
| Inspected Result | Summary | Primary Components | Secondary Components | Unused Components |
| Overall Test Case | Verifies each radar mode returns the correct information | RadarMode | RadarProcessor | N/A |
| | | | | UNCLASSIFIED |

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Table 5: (U) Test Case Prerequisites (Continued)

| UNCLASSIFIED | | | | |
|---|---|---|---|---|
| Inspected Result | Summary | Primary Components | Secondary Components | Unused Components |
| Search Mode | Verifies that Search Mode properly scans and creates virtual tracks on adequate detections | SearchMode | RadarMode RadarProcessor | N/A |
| Acquire Mode | Verifies that Acquire Mode revisits a virtual track and properly converts it into a confirmed track | AcquireMode | RadarMode RadarProcessor | N/A |
| Track Mode | Verifies that Track Mode properly revisits and updates a stable track | TrackMode | RadarMode RadarProcessor | N/A |
| | | | | UNCLASSIFIED |

#### 4.2.3.3 (U) Test Inputs

(U) TIMELINE: Should be finalized by CDR.
(U) REQUIREMENT Identify the inputs for this test that differ from the list of input options in this Test Family's Default Parameters table. Also describe the geometry of the test; a figure in addition to text is strongly encouraged.
(U) EXAMPLE: (U) This test can be ran by executing the TF_1.1.scn file in the Scenario GUI. The inputs for this test are the model inputs listed in the Basic Test Parameters for this test family except for the following changes:

Table 6: (U) Test Case 1X: Unique Parameters

| UNCLASSIFIED | | |
|---|---|---|
| Input Name | Description | Value(s) |
| Simulation Start Date | Start Date for the Simulation | 07/10/2024 |
| | | UNCLASSIFIED |

#### 4.2.3.4 (U) Assumptions and Constraints

(U) TIMELINE: Should be started by CDR and finalized by TRR.
(U) Please include any relevant assumptions and/or constraints that are specific to this test scenario. Some tests will have both assumptions and constraints. In this context, **Assumptions** enumerate which functionalities of the overall model behave differently during this particular test. These include which components are operational in the model (excluding prerequisites), what is working which components are operational in the target, what is working which components are operational in in the ODESSA framework, and what the changes to the active scenario do to the above. Examples: if the system is put into mode X it will require a detection to begin jamming, if the pedestal is static then the antenna pattern will be probed a reasonable amount, if the TAR is disabled the TER will have difficulty finding targets, etc. **Constraints** are restrictions that the scenario has put upon the test framework as a whole, either to change model behavior for testing or to the scenario itself as a whole to limit model behavior. Some scenarios will do both of these. Examples: pedestal rotation is disabled to allow for antenna testing, disabling TAR and using perfect cue to test TER track capabilities, placing a target behind terrain to make it undetectable, etc.

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#### 4.2.3.5 (U) Test Procedure

(U) TIMELINE: Should be started by PDR and finalized by TRR.
(U) REQUIREMENT Identify the required outputs and how the data will be presented. If specific outputs are logged, explain the log location and name, and an explanation of the file format and fields. The explanation here should be brief and lean on the description in the User Manual; if the explanation becomes long, expand the UAM description and shorten the description here. If the data will be graphed, explain what will be plotted and the axis labels, etc., and examples of those figures, tables, etc. should be captured for review and discussion with SMEs and M&S TPOCs in this document and the milestone slides no later than CDR.
(U) Include the filenames of the scenario files and plot configuration files developed to execute this test as applicable.
(U) If an external post-processing tool is used (such as a MATrix LABoratory (MATLAB) sidecar), then add another table and include the default parameters for that, too.
(U) The purpose of this section is so that the SME and TPOC can provide meaningful feedback to the tester early in the process – before the test is run and results presented. An example of feedback that this will enable is the SME could say that "screen shots of log files are meaningless", or "a plot of OBSERVABLE vs. QUANTITY only provides value of you include a second plot with the relative difference calculated", or "that figure should also include truth data", etc. A secondary purpose of this section is that it provides guidance to more junior technicians or reviewers if they are tasked with implementing the test scripts.
It is not sufficient to simply state, "run the test script and then run the data plotting script", or "load the scenario and run it", or "the output is in a log file, plot it". Give specific instructions for recreating any graphs shown here.
(U) EXAMPLE: Figure 1 is generated by plotting BSE azimuth angle vs. time to show a successful track and following of the target, and is also generated by the ST_1.1_fig_1.config file.

#### 4.2.3.6 (U) Expected Results and Evaluation Criteria

(U) TIMELINE: Should be started by PDR and finalized by TRR.
(U) REQUIREMENT Describe the expected results from the test. If this is based on intel instead of physics and engineering, cite the supporting source. Explain what we would expect to see from each of the figures and logged data defined in the Test Procedure section. Note that each step in the test will need its own expected results and disposition assessment in the Result Narrative table, therefore identify a range of accuracy, allowable errors, etc. to assist the tester in determining the disposition of each testing step. If provided intelligence predicts a particular result, then that result should be called out in the expected results. Show math to bolster the expectation and potentially describe the output from the model that will support the calculations. Describe which KPIs are assessed by this test and whether this test and/or KPI can be used for validation.
(U) Similar to the test procedure section, this section provides the most value early in the development and documentation process. This section allows the SME to provide feedback on whether or not the test case has appropriate parameter coverage or will provide sufficient and unambiguous evidence. A secondary purpose of this section is that it provides guidance to more junior technicians or reviewers if they are tasked with implementing the test scripts and reviewing results.
(U) Note: it is not sufficient to simply state, "we expect it to behave correctly." Give specific benchmarks for success or failure. For example, "If the pedestal orientation versus time does not settle to the pedestal command versus time with a specified time, the test fails."
(U) For example, "As A and B move apart, we expect the communication connection to match that predicted in equation 1. Based on the characteristics of this system, we expect loss of connection to occur at an MDS below YYY; and based on the setup of this scenario we expect this to occur at a range of XXX meters. Varying parameter CC should result in a linear increase in observable DD for the range FFF to GGG."
(U) Correlate each of the items in Behavior Tested above with an expected result. Quantify the expected results including an acceptable error margin wherever possible.

#### 4.2.3.7 (U) Result Narrative

(U) TIMELINE: Should be started by CDR near-finalized by TRR, and finalized at FDR.
(U) REQUIREMENT Give a narrative description of the actual result from executing the test case. Ensure that any deviations from the expected results are described in enough detail to assist in analyzing the results. Call out any unique features and explain them. Assess whether the test is PASS or FAIL. Refer to the generated figures. Figures, tables, charts, etc. used to verify model performance should be review with SMEs and M&S TPOCs starting no later than CDR and should be refined and solidified by TRR. In general, screen shots of log output is not acceptable.

(U) Test Narrative Goes Here

(U) TIMELINE: Should be started by CDR near-finalized by TRR, and only minor variations in test results should occur between TRR and FDR.
(U) Document any minor variations on this test in the below table, if applicable. These tests should only be small 1-2 parameter changes that do not drastically affect testing results. If using batch run permutations, separate the input parameters with ; as in the Scenario GUI. If using multiple scenario files, make a separate line for each scenario file.

Table 7: (U) Test Case Permutations

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| Scenario File | Description | Varied Parameters | Expected Changes | Test Disposition |
|---|---|---|---|---|
| 1.2b.scn | moved target slightly North | STL Latitude = 0.41 deg | System should still be able to track the target | **PASS** |
| 1.2c.scn | moved target slightly East and West | STL Longitude = 0.41;-0.41 deg | System should still be able to track the target | **PASS** |

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(U) TIMELINE: Should be started by CDR near-finalized by TRR, and finalized at FDR.
(U) REQUIREMENT Use the table below to document the steps for the test case.
Step #: the test case step number.Each step number will tie to individual steps to the final complete test. Multiple steps may tie to individual requirements that are demonstrated through a single test.
Description: the required input by the tester (tester actions and equipment operation) that is required in order to achieve the expected result
Expected Results: the expected results from performing the step. Identify a range of accuracy, allowable errors, etc. that will assist the tester in determining the disposition of the step.
Disposition: the disposition of the step
Actual Results: the actual result from executing the test case. Ensure that any deviations from the expected results are described in enough detail to assist in analyzing the results.
Comments: include comments as appropriate
Test Case Disposition: the disposition of the test PASS, PASS with CR, FAIL, PENDING

Table 8: (U) Test Case SampleTestCaseID: SampleTestCaseName

| UNCLASSIFIED | | | |
|---|---|---|---|
| Step # | Description | Expected Results | Disposition |
| 1 | description | expected results | PASS |
| Actual Results | | Comments: | |
| your text of the actual results | | your comments | |
| Test Case Disposition | | **PASS** | |
| | | | **UNCLASSIFIED** |

# 5 (U) Unit Tests

(U) TIMELINE: Should be started by PDR and finalized at TRR.
(U) Describe any unit tests that are used to verify requirements are met. Make sure that the unit tests listed in the Section 3 tie to the unit test number listed here.
(U) This section documents the unit tests written for this model. Unit tests are small, focused tests that verify the correctness of individual components or functions in isolation. Their primary goal is to ensure that the code behaves as expected under various conditions and inputs.
(U) The tests in this project are implemented using the Google Test framework. To evaluate test coverage and identify untested branches or paths, the project uses OpenCppCoverage.
(U) Test cases include a description, parameter information, and a breakdown of expected behaviors grouped by test stages, where applicable.

Table 9: (U) Unit Tests for Requirements

| UNCLASSIFIED | | | |
|---|---|---|---|
| Requirement | Description | Unit Test Number | Test Result / Status |
| 9.1 | Description of unit test | UT1 | Passed |
| | | | **UNCLASSIFIED** |

# 6 (U) Sources

(U) Reference the UAM for a comprehensive list of sources used to develop this model. Reference Citations below for those sources used in creating this TDD.

# 7 (U) Acronyms

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| Term | Description | Page |
|---|---|---|
| BAH | *Booz Allen Hamilton* | 1 |
| CDR | *Critical Design Review* | 6, 7, 8, 9 |
| EWR | *Early Warning Receiver* | 7 |
| FDR | *Final Design Review* | 2 |
| KPI | *Key Performance Indicator* | 4 |
| MATLAB | *MATrix LABoratory* | 10 |
| NGIC | *National Ground Intelligence Center* | 1, 2, 6 |
| ODESSA | *Overarching Dynamic Electronic-Warfare System Standard Architecture* | 1 |
| TDD | *Test Description Document* | 1, 12 |
| TMAP | *Threat Modeling and Analysis Program* | 1 |
| TPOC | *Technical Point of Contact* | 2 |
| TRR | *Test Readiness Review* | 2, 9 |
| UAM | *User and Analyst Manual* | 4, 12 |

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